MATH0076 Algebraic Geometry

Year:	2026–2027
Code:	MATH0076
Level:	7 (UG), 7 (PG)
Normal student group(s):	UG: Y4 Mathematics programmes PG: MSc Mathematics
Value:	15 credits (7.5 ECTS credits)
Term:	2
Assessment:	90% examination, 10% coursework
Normal Pre-requisites:	Background in algebra and topology
Lecturer:	Prof A Yafaev

Course Description and Objectives

Algebraic Geometry is the study of algebraic varieties, spaces which are defined by polynomial equations in several variables. Although the subject goes back to Descartes, it is still one of the most thriving research areas of pure mathematics. It is intimately connected to commutative algebra, and closely linked to many other areas of mathematics including number theory, differential geometry, and representation theory

Our aim is to introduce basic notions of algebraic geometry, using lots of explicit examples

Recommended Texts

- R Hartshorne, *Algebraic Geometry*
- W Fulton, *Algebraic curves*
- M Reid, *Undergraduate algebraic geometry*
- IR Shafarevich, *Basic algebraic geometry*

Detailed Syllabus

- Affine algebraic varieties: definitions, basic properties.
- Tangent spaces, singularities, dimension.
- Morphisms, rational and birational maps.
- Projective varieties.
- Blow-ups.
- Divisors. Bezout's theorem.